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Lab Sheet-03

Lab Assessment on Supervised Learning (Regression Models)

MCA III Semester (Session 2026–2027)

Objective:

- To understand the fundamentals of supervised learning using regression techniques.
- To implement Linear Regression and Polynomial Regression models.
- To train and test regression models using real-world datasets.
- To evaluate regression model performance using various evaluation metrics.
- To compare regression models based on prediction accuracy.

Requirements:

- Python 3.11 or above
- Jupyter Notebook / JupyterLab / VS Code / PyCharm
- NumPy
- Pandas
- Matplotlib
- Seaborn
- Scikit-learn
- Sample Regression Dataset (Housing Price, Salary, Advertising, California Housing, etc.)

Program List

1. Load a regression dataset using the Pandas library.
2. Display the first and last five records of the dataset.
3. Explore dataset information and descriptive statistics.
4. Identify input (independent) and output (dependent) variables.
5. Split the dataset into training and testing sets.
6. Implement a Simple Linear Regression model.
7. Train the Linear Regression model using the training dataset.
8. Predict output values using the trained Linear Regression model.
9. Visualize the Linear Regression line using Matplotlib.

10. Compare actual and predicted values.
11. Display the regression coefficient and intercept.
12. Predict the output for new user-defined input values.
13. Implement Multiple Linear Regression using multiple independent variables.
14. Train the Multiple Linear Regression model.
15. Predict output values using the testing dataset.
16. Compare actual and predicted values graphically.
17. Analyze the effect of each independent variable on prediction.
18. Implement Polynomial Regression of Degree 2.
19. Implement Polynomial Regression of Degree 3.
20. Compare Linear Regression and Polynomial Regression models.
21. Visualize Polynomial Regression curves.
22. Predict output values using the Polynomial Regression model.
23. Compare prediction accuracy for different polynomial degrees.

Regression Model Evaluation

24. Calculate Mean Absolute Error (MAE).
25. Calculate Mean Squared Error (MSE).
26. Calculate Root Mean Squared Error (RMSE).
27. Calculate the R-squared (R^2) Score.
28. Compare the performance of Linear and Polynomial Regression using evaluation metrics.
29. Interpret the meaning of MSE and R^2 values.
30. Visualize prediction errors using scatter plots.

Model Improvement and Analysis

31. Train the regression model using standardized features.
32. Compare model performance before and after feature scaling.
33. Perform regression using another real-world dataset.
34. Save the trained regression model using the Joblib library.
35. Load the saved model and use it to predict new data.

Instructions:

- Write comments to make your programs readable.

- Use meaningful and descriptive variable names.
- Display all intermediate and final outputs clearly.
- Visualize regression models wherever applicable.
- Use appropriate evaluation metrics for every model.
- Save trained models after successful implementation.

Guidelines:

- Follow Python coding standards (PEP-8).
- Use Pandas and NumPy for data preprocessing.
- Use Matplotlib and Seaborn for data visualization.
- Use Scikit-learn for implementing regression algorithms.
- Split the dataset into training and testing sets before model training.
- Interpret the results obtained from evaluation metrics.
- Document each experiment using Markdown cells in Jupyter Notebook.

Submission:

- Submit the source code files (.ipynb or .py) for each experiment.
- Include screenshots of successful execution and generated graphs.
- Attach a brief report containing:
    - **Objective**
    - **Algorithm/Procedure**
    - **Source Code**
    - **Output**
    - **Performance Metrics (MAE, MSE, RMSE, R^2)**
    - **Observation**
    - **Conclusion**
- Submit the dataset(s) used in the experiments.
- Ensure all required Python libraries are installed and documented before submission.